

CHENG, Keguang

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Research interests: Evolutionary biology, plant genomics, ecology and biodiversity conservation.

EDUCATION

New York University (NYU)

New York, NY, USA

M.S., Biology (General Biology), expected May 2027 | GPA: 3.889/4.0

Selected Coursework: Machine Learning in Medicine and Biology; Biological Databases and Data-Mining; Applied Genomics; Statistics in Biology; Programming for Biologists.

Beijing Forestry University (BJFU)

Beijing, China

B.S., Biological Sciences (National Science Base), Jul 2024 | GPA: 3.75/4.0

PUBLICATIONS

- Jing, Z.-Y., Zhang, R.-G., Liu, Y., **Cheng, K.-G.**, Liu, D.-T., Shu, H., Kong, J., Liu, Z.-H., Ma, Y.-P., & Liu, P.-L. (2025). Genomic insights into the evolutionary history and conservation of the living fossil *Tetracentron sinense*. *Plant Diversity*, 47(5), 759–771. doi:10.1016/j.pld.2025.05.008
- Jing, Z.-Y., **Cheng, K.-G.**, Shu, H., Ma, Y.-P., & Liu, P.-L. (2023). Whole genome resequencing approach for conservation biology of endangered plants. *Biodiversity Science*, 31(5), 22679. doi:10.17520/biods.2022679 [Review article; in Chinese]

MANUSCRIPTS IN PREPARATION

- Beyond Canonical NLR Classes: Phylogenetic Diversity and Chromosomal Organization in Gymnosperms. *First-author research manuscript; working title.*
- From Evolution to Design: Diversity, Constraints, and Engineering of Plant NLR Immune Receptors. *First-author review manuscript.*

RESEARCH EXPERIENCE

Research Project | Organ identity and development in *Podostemum ceratophyllum*

New York Botanical Garden (NYBG) | Advisor: Cecilia Zumajo

May 2026–Present

- Investigating the developmental basis of unusual organ morphology through comparative analysis of root, leaf and whole-plant RNA-seq data.
- Developed the literature review and computational analysis plan; used Trinity-based de novo assembly and transcriptome quality assessment to establish a reference for developmental gene-expression analysis.
- Screened developmental regulators using reciprocal BLAST to guide planned phylogenetic validation and root–leaf expression comparisons aimed at resolving organ homology.

Independent Researcher | NLR gene-family evolution in gymnosperms

BJFU | Advisor: Pingli Liu

Mar 2023–Present

- Classified NLRs across 29 gymnosperm genomes by domain architecture; reconstructed phylogenies under alternative classifications, addressing long-branch attraction and incorporating angiosperm and algal sequences.
- Resolved a large, strongly supported lineage outside the canonical TNL, RNL and CNL classes, positioned between TNL and non-TNL clades in phylogenetic analyses.
- Mapped NLR loci and gene clusters; observed concentration on a single chromosome in multiple gymnosperm species and tested chromosome-level enrichment.

Team Member | Population genomics of *Tetracentron sinense*

BJFU | Advisor: Pingli Liu

Feb 2022–Mar 2025

- Analyzed population structure from genome-wide resequencing data using PCA and ADMIXTURE to characterize genetic differentiation and ancestry patterns.
- Estimated nucleotide diversity (π), Watterson's θ and population differentiation (F_{ST}) to compare genetic variation within and among populations and inform conservation-genomic interpretation.
- Created, revised and assembled all manuscript figures in R and Adobe Illustrator, integrating analysis outputs into publication-ready visualizations.

Course Project | RNA-seq analysis of NLR expression in *Pinus tabuliformis*

NYU, Applied Genomics | Instructor: Manpreet Katari

Spring 2026

- Reanalyzed a public 12-sample time-course RNA-seq dataset from pine wood nematode-infected *Pinus tabuliformis* using a curated catalogue of 661 NLR genes, Salmon/tximport and DESeq2 likelihood-ratio testing.
- Examined class-specific temporal expression with Mfuzz and WGCNA while accounting for module-size effects; used exploratory co-expression and promoter-motif analyses to prioritize candidate NLRs and transcription-factor families.
- Documented the workflow in R Markdown and produced figures, tables and an interactive Shiny dashboard.

Team Member | Functional analysis of very small introns in plants

BJFU | Advisor: Hongbo Gao

Dec 2022–Jun 2023

- Reviewed intron splicing and evolution; performed primer design, RT-PCR and gel electrophoresis to investigate minimum intron length and splicing accuracy, and used Python for data processing and figure preparation.

Team Lead | Cold-stress physiology in Daurian ground squirrel

BJFU | Advisor: Qiang Weng

Sep 2021–May 2022

- Led a four-member team studying cold-induced mitochondrial adaptation; designed assays for mitochondrial abundance and oxidative-stress markers, coordinated troubleshooting, and presented results in a team report and presentation.

TECHNICAL SKILLS AND TRAINING

- **Computing and HPC:** Linux/Bash; Slurm job arrays; Git; Jupyter, R Markdown and Shiny.
- **Programming and data:** Python (NumPy, pandas, Biopython); R (tidyverse, ggplot2); SQL (MySQL, SQLite); Adobe Illustrator.
- **Genomics:** RNA-seq quality control and differential expression; de novo transcriptome assembly and evaluation; gene-family analysis; phylogenetics; population genomics; VCF processing.
- **Statistical training:** hypothesis testing; linear and logistic regression; ANOVA; PCA; bootstrap and permutation tests; false discovery rate control.
- **Machine learning:** scikit-learn; classification; cross-validation.
- **Wet lab:** RT-PCR; gel electrophoresis; fluorescence staining; Western blot.

ADDITIONAL TRAINING

Fudan University | Shanghai, China

Jul–Aug 2025

Graduate International Summer School of Life Sciences: Genetics and Developmental Biology (Jul); Evolution and Biological Big Data (Aug).

FIELD EXPERIENCE

- Ecology field internship (Jul 2023): conducted forest surveys and led a 10-member team studying slope aspect and vegetation indices.
- Field surveys and practicums (2021): conducted medicinal-plant surveys, botanical and zoological specimen collection and identification, and a salinity-stress experiment.

HONORS

- BJFU Academic Excellence Scholarship (2022, 2023); BJFU Outstanding Student Scholarship (2021, 2023).
- Second Prize, 9th Beijing University Student Biology Knowledge Competition (2023).
- Third Prize, Beijing University Student Biology Experimental Design Competition (2022, 2023).